

QUBO Formulation: Quantum-Circuit Qubit Allocation

Quantum-Circuit Qubit Allocation

Problem Statement. Given a set of logical qubits L , a set of physical qubits P , and a hardware coupling graph that defines the connectivity and fidelity between physical qubits, the task is to assign each logical qubit to a distinct physical qubit. Let $G \subseteq L \times L$ denote the set of logical qubit pairs that participate in two-qubit gates, and let $C_{p,q}$ denote the penalty cost associated with mapping a logical pair to physical qubits p and q . The goal is to find a one-to-one mapping that minimizes the total penalty incurred by two-qubit gates mapped onto distant or low-fidelity physical pairs.

Decision Variables. For each logical qubit $l \in L$ and physical qubit $p \in P$, we define a binary decision variable $x_{l,p} \in \{0, 1\}$. The variable $x_{l,p}$ equals 1 if logical qubit l is mapped to physical qubit p , and 0 otherwise. The total number of variables is $|L| \times |P|$.

QUBO Derivation. Step 1 — Objective Function. The objective is to minimize the total penalty for two-qubit gates mapped to physical pairs. This is expressed as:

$$\min_x J(x) = \sum_{(l_i, l_j) \in G} \sum_{p \in P} \sum_{q \in P} C_{p,q} x_{l_i,p} x_{l_j,q}. \quad (1)$$

Step 2 — Constraints. The mapping must satisfy two fundamental constraints. First, each logical qubit must be assigned to exactly one physical qubit:

$$\sum_{p \in P} x_{l,p} = 1, \quad \forall l \in L. \quad (2)$$

Second, each physical qubit can host at most one logical qubit:

$$\sum_{l \in L} x_{l,p} \leq 1, \quad \forall p \in P. \quad (3)$$

Step 3 — Penalty Term Conversion. We convert the constraints into quadratic penalty terms added to the objective. Let $\lambda > 0$ be a penalty weight.

(a) For the logical assignment constraint, we introduce:

$$H_{C1}(x) = \lambda \sum_{l \in L} \left(\sum_{p \in P} x_{l,p} - 1 \right)^2. \quad (4)$$

(b) Expanding the square and applying the idempotent property $x_{l,p}^2 = x_{l,p}$ for binary variables yields:

$$\begin{aligned} \left(\sum_{p \in P} x_{l,p} - 1 \right)^2 &= \sum_{p \in P} x_{l,p}^2 - 2 \sum_{p \in P} x_{l,p} + 1 \\ &= \sum_{p \in P} x_{l,p} - 2 \sum_{p \in P} x_{l,p} + 1 \\ &= 1 - \sum_{p \in P} x_{l,p}. \end{aligned} \quad (5)$$

(c) The resulting general penalty contribution is:

$$H_{C1}(x) = \lambda \sum_{l \in L} \left(1 - \sum_{p \in P} x_{l,p} \right) = \lambda n_L - \lambda \sum_{l \in L} \sum_{p \in P} x_{l,p}. \quad (6)$$

For the physical capacity constraint, we penalize any assignment where two or more logical qubits occupy the same physical qubit:

$$H_{C2}(x) = \lambda \sum_{p \in P} \left(\sum_{l \in L} x_{l,p} \right)^2. \quad (7)$$

Expanding and applying idempotence:

$$\begin{aligned} \left(\sum_{l \in L} x_{l,p} \right)^2 &= \sum_{l \in L} x_{l,p}^2 + \sum_{l \in L} \sum_{l' \in L, l' \neq l} x_{l,p} x_{l',p} \\ &= \sum_{l \in L} x_{l,p} + \sum_{l \neq l' \in L} x_{l,p} x_{l',p}. \end{aligned} \quad (8)$$

The resulting penalty contribution is:

$$H_{C2}(x) = \lambda \sum_{p \in P} \left(\sum_{l \in L} x_{l,p} + \sum_{l \neq l' \in L} x_{l,p} x_{l',p} \right). \quad (9)$$

Step 4 — Combined QUBO Hamiltonian. Summing the objective and penalty terms gives the total Hamiltonian $H(x)$:

$$H(x) = \sum_{(l_i, l_j) \in G} \sum_{p \in P} \sum_{q \in P} C_{p,q} x_{l_i,p} x_{l_j,q} + \lambda \sum_{l \in L} \sum_{p \in P} x_{l,p} + \lambda \sum_{p \in P} \sum_{l \neq l' \in L} x_{l,p} x_{l',p} + \lambda n_L. \quad (10)$$

Note that the linear terms from H_{C1} and H_{C2} cancel exactly, leaving only quadratic and constant terms.

Step 5 — Q Matrix Entries and Offset. Reading off the coefficients from $H(x) = \sum_{i \leq j} Q_{ij} x_i x_j + c$, we obtain the following closed-form expressions for the QUBO matrix entries and offset:

$$Q_{(l_i,p),(l_j,q)} = C_{p,q} \quad \text{if } (l_i, l_j) \in G \text{ and } (l_i, p) \neq (l_j, q), \quad (11)$$

$$Q_{(l,p),(l',p)} = \lambda \quad \text{if } l \neq l', \quad (12)$$

$$Q_{(l,p),(l,p)} = 0 \quad \text{for all } l \in L, p \in P, \quad (13)$$

$$c = \lambda n_L. \quad (14)$$

Penalty Weight Justification. The penalty weight λ must be chosen such that any constraint violation incurs a cost strictly greater than the maximum possible reduction in the objective function. The objective function is bounded by $|G| \cdot \max_{p,q} |C_{p,q}|$. Therefore, setting $\lambda > |G| \cdot \max_{p,q} |C_{p,q}|$ guarantees that the energy of any infeasible configuration exceeds that of the optimal feasible configuration, ensuring that the optimizer prioritizes feasibility over objective minimization.

Correctness Argument. Minimizing $H(x)$ over the binary hypercube $\{0, 1\}^{|L| \times |P|}$ recovers the optimal qubit allocation because feasible mappings satisfy all constraints, yielding zero penalty terms and reducing $H(x)$ to the original objective $J(x)$. Conversely, any infeasible assignment violates at least one constraint, incurring a penalty strictly larger than the maximum possible objective gain from that violation. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment with the minimum allocation cost.

Illustrative Example. Consider a small instance with $n_L = 2$ logical qubits ($L = \{0, 1\}$) and $n_P = 2$ physical qubits ($P = \{0, 1\}$). Suppose there is a single two-qubit gate between logical qubits 0 and 1, so $G = \{(0, 1)\}$. The variable mapping is $x_{0,0} \rightarrow x_0$, $x_{1,0} \rightarrow x_1$, $x_{0,1} \rightarrow x_2$, $x_{1,1} \rightarrow x_3$. Using the general formulas with penalty weight $\lambda = 100$ and offset $c = 200$, the objective terms are:

$$H_{\text{obj}} = C_{0,0}x_0x_1 + C_{0,1}x_0x_3 + C_{1,0}x_1x_2 + C_{1,1}x_2x_3. \quad (15)$$

The penalty terms expand to:

$$H_{C1} = 100(1 - x_0 - x_2) + 100(1 - x_1 - x_3) = 200 - 100(x_0 + x_1 + x_2 + x_3), \quad (16)$$

$$H_{C2} = 100(x_0x_2 + x_1x_3) + 100(x_0 + x_2) + 100(x_1 + x_3). \quad (17)$$

Combining these, the linear terms cancel, yielding the full Hamiltonian:

$$H(x) = C_{0,0}x_0x_1 + C_{0,1}x_0x_3 + C_{1,0}x_1x_2 + C_{1,1}x_2x_3 + 100x_0x_2 + 100x_1x_3 + 200. \quad (18)$$

The optimal mapping assigns logical qubit 0 to physical qubit 0 and logical qubit 1 to physical qubit 1 (i.e., $x_0 = 1$, $x_3 = 1$, others 0), yielding an energy of $C_{0,1} + 200$. Alternatively, mapping

$0 \rightarrow 1$ and $1 \rightarrow 0$ gives $C_{1,0} + 200$. The minimum energy corresponds to the physical pair with the lowest gate penalty, demonstrating how the QUBO formulation correctly encodes the allocation task.